Discrete Stochastic Cascade Dynamics on Finite Graphs with Irreversible Energy Depletion

Part I: Construction, Guarantees, and Phase Structure

Shiv Goswami

Terraflock Pvt Ltd, Bihar, India

https://shivgoswami.com | https://github.com/Shiv2071/Discrete-cascade-model

March 23, 2026

Abstract

We construct a new class of discrete stochastic dynamical systems on finite graphs that generate rich, transient structure from irreversible resource consumption alone. Two fundamentally asymmetric excitation species, distinguished by intrinsic frequency and self-interaction rules, interact stochastically while drawing from a strictly non-renewable local energy field. Four coupled state variables (excitation counts, capacity energy, structural state, and a ripple measure) evolve through layered update rules that produce three dynamical regimes: quiescent, dissipative leakage, and explosive pair creation. A Landau-type free-energy functional governs bond formation, embedding a genuine order–disorder phase transition within the cascade dynamics and giving rise to a four-level structural hierarchy (excitation $\rightarrow$ pair $\rightarrow$ bond $\rightarrow$ clump).

We establish three rigorous guarantees. *Energy Monotonicity:* the total capacity energy is a non-negative supermartingale that decreases strictly in expectation whenever the system is active, providing a built-in thermodynamic arrow. *Finite Activity Bound:* the cumulative number of interactions across all space and time is almost surely finite, bounded by the ratio of initial energy to unit interaction cost. *Almost Sure Absorption:* the system reaches a frozen absorbing configuration in finite time with probability one, without requiring fine-tuning to a critical point.

Beyond these guarantees, we identify and characterise a transient complexity peak: a regime in which structural diversity first increases through explosive cascades before decaying as energy is exhausted. We derive a complete phase diagram over initial energy density and interaction strength. The resulting framework demonstrates that spatial structure, self-amplifying cascades, metastable bonded configurations, and guaranteed finite-time termination can all emerge from purely discrete rules with no continuum limit, no external drive, and no infinite-resource assumption.

Keywords: interacting particle systems; absorbing-state phase transitions; stochastic cascades; irreversible dynamics; discrete dynamical systems; Landau phase transition

Contents

1 Introduction

1.1 Motivation

The standard mathematical description of motion rests on a chain of continuum assumptions: space is $\mathbb{R}^d$, time is $\mathbb{R}$, trajectories are smooth (or at least piecewise differentiable), and evolution is governed by differential equations whose solutions require limits over infinitely fine subdivision. This framework, formalised through calculus, has been extraordinarily successful. Yet it carries a foundational dependence on the completed infinite: every derivative is a limit, every integral an infinite sum, and every trajectory the result of infinitely many infinitesimal steps.

This dependence is not merely aesthetic. Zeno's dichotomy, that an object traversing a distance must first traverse half, then half the remainder, *ad infinitum*, is resolved in calculus by the convergence of geometric series, but the resolution presupposes exactly the continuum structure that the paradox questions. If one asks not "does the series converge?" but "can we build dynamics that do not *require* infinite subdivision?", a different programme emerges: replace smooth trajectories with discrete evolution rules, replace fields on $\mathbb{R}^d$ with states on finite graphs, and replace conservation laws with irreversible resource budgets.

The goal is not to refute calculus, whose empirical adequacy is beyond question, but to construct a self-contained dynamical framework in which:

(i) space is a finite graph with no notion of subdivision,

(ii) evolution proceeds in discrete steps with no limiting process,

(iii) two asymmetric excitation species interact stochastically,

(iv) a strictly non-renewable capacity energy imposes irreversibility, and

(v) threshold-driven regime switching generates qualitative diversity from quantitative variation.

The result is a minimal system that generates rich transient dynamics, including spatial structure, self-amplifying cascades, and metastable bonded configurations, yet is guaranteed to freeze into an absorbing state in finite time. Motion, geometry, and thermodynamic direction emerge from the interplay of discrete rules rather than from continuum assumptions.

1.2 Related Work

The model sits at the intersection of several classical threads.

Interacting particle systems. The contact process [Harris, 1974, Liggett, 1999] and voter model study binary-state agents on lattices with local stochastic transitions. Our model extends this to multi-species, multi-field dynamics with a global energy constraint.

Absorbing-state phase transitions. Directed percolation universality [Hinrichsen, 2000, Henkel et al., 2008] governs phase transitions into absorbing configurations in many one-component systems. Our model introduces a *guaranteed* absorbing transition via irreversible energy depletion, removing the need for fine-tuning to a critical point.

Self-organised criticality. Sandpile models [Bak et al., 1987, Dhar, 1990] exhibit power-law avalanche distributions through slow external driving balanced by fast relaxation. In our model there is *no external drive*: the system runs on a finite initial endowment, and the "self-organised" behaviour is a transient phenomenon.

Reaction–diffusion systems. Continuous-space models of two-species annihilation ($A + B \to \varnothing$) and branching ($A \to 2A$) exhibit rich kinetics [Toussaint & Wilczek, 1983, Lee, 1994]. Our framework discretises and couples these with energy budgets and structural memory.

1.3 Contributions

This paper makes four contributions:

1. **A new dynamical system class.** We define a fully discrete, finite-resource stochastic system on arbitrary finite graphs, coupling four state variables through three dynamical regimes. The system requires no continuum limit, no external drive, and no infinite-resource assumption (section 2).

2. **Three rigorous guarantees.** We prove that energy decreases monotonically, that total activity is almost surely finite, and that the system absorbs in finite time with probability one. Together, these establish an intrinsic thermodynamic arrow and guaranteed termination (section 3).

3. **Embedded phase transition.** A Landau free-energy functional governs bond formation, embedding a genuine local order–disorder transition within the cascade dynamics and producing a four-level structural hierarchy (section 2.9).

4. **Phase diagram and complexity peak.** We derive a complete phase diagram over initial energy density and interaction strength, and characterise a transient complexity peak where structural diversity first rises through explosive cascades before decaying toward absorption (section 4).

1.4 Organisation

Section 2 defines the model. Section 3 states and proves the main theorems. Section 4 analyses the phase structure. Section 5 discusses emergent properties and interpretation. Section 6 places the work in context and identifies open problems.

2 Model Definition

2.1 Graph Topology

Definition 2.1 (Spatial substrate)**.** Let $\mathcal{G} = (\mathcal{V}, \mathcal{A})$ be a finite, connected, undirected graph with vertex set $\mathcal{V}$, $|\mathcal{V}| = P < \infty$, and adjacency relation $\mathcal{A} \subseteq \mathcal{V} \times \mathcal{V}$. For each vertex $p \in \mathcal{V}$ the *neighbourhood* is $\mathcal{N}(p) = \{q \in \mathcal{V} : (p, q) \in \mathcal{A}\}$, with degree $\deg(p) = |\mathcal{N}(p)|$.

Remark 2.2. The graph $\mathcal{G}$ may be a regular lattice ($\mathbb{Z}^d$ with periodic boundary conditions), a random graph, or any finite connected graph. Results in section 3 hold for arbitrary $\mathcal{G}$.

2.2 State Variables

At each vertex $p \in \mathcal{V}$ and discrete time step $n \in \mathbb{N}_0$, the system carries four state variables:

Definition 2.3 (Local state)**.** The *state* at vertex p and time n is the tuple

$$\sigma(p, n) = \big(X(p, n),\ Y(p, n),\ \mathcal{E}(p, n),\ S(p, n)\big) \ \in\ \mathbb{N}_0 \times \mathbb{N}_0 \times \mathbb{R}_{\geq 0} \times \mathbb{R}\,,$$

where:

- $X(p, n) \in \mathbb{N}_0$: count of *type-X excitations* (species 1), with intrinsic frequency $\omega_X > 0$,
- $Y(p, n) \in \mathbb{N}_0$: count of *type-Y excitations* (species 2), with intrinsic frequency $\omega_Y > 0$, $\omega_Y \neq \omega_X$,
- $\mathcal{E}(p, n) \in \mathbb{R}_{\geq 0}$: *local capacity energy*,
- $S(p, n) \in \mathbb{R}$: *structural state*, encoding accumulated interaction history and bonded configurations.

The two species are *fundamentally asymmetric*: they carry different intrinsic frequencies, obey different self-interaction rules (see section 2.4), and are not interchangeable under any symmetry of the model.

The *global state* at time n is the collection $\Sigma(n) = \{\sigma(p, n)\}_{p \in \mathcal{V}}$.

Definition 2.4 (Total capacity energy)**.**

$$\mathcal{E}_{\text{tot}}(n) \ =\ \sum_{p \in \mathcal{V}} \mathcal{E}(p, n)\,.$$

2.3 Parameters

The model is governed by the following constants:

Symbol	Name	Constraint
ω_X, ω_Y	Intrinsic frequencies	$\omega_X, \omega_Y > 0, \omega_X \neq \omega_Y$
α_{XY}	Cross-interaction coefficient	$\alpha_{XY} > 0$
α_{XX}	X self-interaction coefficient	$\alpha_{XX} \geq 0$
k_{XY}	Energy cost per XY interaction	$k_{XY} > 0$
k_{XX}	Energy cost per XX interaction	$k_{XX} > 0$
C	Ripple threshold	$C > 0$
Δ	Overshoot margin	$\Delta > 0$
λ	Leakage rate	$\lambda > 0$
D_X, D_Y	Diffusion coefficients	$D_X, D_Y \in [0, 1/\deg_{\max})$
γ_1, γ_2	Structural coupling	$\gamma_1, \gamma_2 > 0$
γ_{XX}	XX structural coupling	$\gamma_{XX} \geq 0$
a_0	Free-energy linear coefficient	$a_0 > 0$
b	Free-energy quartic stabiliser	$b > 0$
T_c	Critical temperature for bonding	$T_c > 0$
κ	Energy cost per bond	$\kappa > 0$
η	Energy cost per explosion	$\eta > 0$

2.4 Interaction Dynamics

The two species obey *asymmetric* interaction rules: X excitations interact both with Y excitations (cross-interaction) and with other X excitations (self-interaction), while Y excitations interact *only* with X. The Y–Y channel is forbidden.

Definition 2.5 (Interaction rules and asymmetry)**.** At each vertex p and time step n, three interaction channels are evaluated:

(i) Cross-interaction (X–Y). The expected cross-interaction rate is

$$R_{XY}(p,n) = \alpha_{XY}\,\omega_X\,\omega_Y\,X(p,n)\,Y(p,n)\,.$$

The realised count is $N_{XY}(p,n) \sim \mathrm{Poi}\big(R_{XY}(p,n)\big)$, capped at $\min\{X(p,n),\ Y(p,n)\}$. Each cross-interaction annihilates one X and one Y:

$$X(p,n) \ \leftarrow\ X(p,n) - N_{XY}(p,n)\,, \tag{1}$$

$$Y(p,n) \ \leftarrow\ Y(p,n) - N_{XY}(p,n)\,. \tag{2}$$

(ii) Self-interaction (X–X). The expected X self-interaction rate is

$$R_{XX}(p,n) = \alpha_{XX}\,\omega_X^2\,\frac{X(p,n)\big(X(p,n)-1\big)}{2}\,,$$

counting unordered pairs. The realised count is $N_{XX}(p,n) \sim \mathrm{Poi}\big(R_{XX}(p,n)\big)$, capped at $\lfloor X(p,n)/2 \rfloor$. Each self-interaction annihilates two X excitations:

$$X(p,n) \ \leftarrow\ X(p,n) - 2\,N_{XX}(p,n)\,. \tag{3}$$

(iii) Y–Y interaction: forbidden. No Y–Y interactions occur:

$$N_{YY}(p,n) = 0 \qquad \text{for all } p,\ n\,. \tag{4}$$

All interaction counts are conditionally independent across vertices and channels given $\Sigma(n)$.

Remark 2.6 (Source of asymmetry)**.** The asymmetry $\alpha_{XX} \geq 0$, $\alpha_{YY} = 0$ is a *constitutive* property of the two species, not a consequence of dynamics. Combined with the distinct intrinsic frequencies $\omega_X \neq \omega_Y$, this makes the species genuinely non-interchangeable: the model has no $X \leftrightarrow Y$ symmetry. This asymmetry is the minimal structural assumption needed for the emergent hierarchy described in section 5.2.

2.5 Energy Depletion

Every interaction irreversibly consumes capacity energy.

Definition 2.7 (Energy cost of interaction)**.** The total energy depleted by interactions at vertex p is

$$\mathcal{C}_{\text{int}}(p,n) = k_{XY} \cdot N_{XY}(p,n) + k_{XX} \cdot N_{XX}(p,n)\,.$$

Remark 2.8. Energy is *consumed by* interactions, not released into the capacity field. The capacity field $\mathcal{E}$ acts as a finite fuel reserve that is irreversibly spent. The distinct costs k_{XY} and k_{XX} allow cross- and self-interactions to have different thermodynamic weights.

2.6 Ripple Measure

Definition 2.9 (Ripple intensity)**.** For $n \geq 2$, define the *ripple intensity* at vertex p as the discrete second-order temporal variation of the structural state:

$$F(p,n) = \left|S(p,n) - 2\,S(p,n{-}1) + S(p,n{-}2)\right|.$$

For $n < 2$, set $F(p,n) = 0$.

Equivalently, $F(p,n) = |\Delta^2 S(p,n)|$, the discrete second difference of S in time. The ripple intensity measures the magnitude of oscillatory change in structural state, a proxy for local dynamical volatility.

2.7 Regime Classification

The ripple intensity classifies each vertex into one of three regimes at each time step.

Definition 2.10 (Dynamical regimes)**.** At vertex p, time n:

- **R1. Quiescent regime:** $F(p,n) \leq C$. No leakage, no explosion. The leakage and explosion contributions are zero: $L(p,n) = 0$, $M(p,n) = 0$.
- **R2. Leakage regime:** $C < F(p,n) < C + \Delta$. Slow dissipation:

  $$L(p,n) = \lambda\, F(p,n)\,.$$

  No explosion: $M(p,n) = 0$.

R3. Explosive regime: $F(p,n) \geq C + \Delta$. An explosion event occurs:

$$m(p,n) = \left\lfloor \frac{F(p,n) - C}{\Delta} \right\rfloor, \tag{5}$$

$$M(p,n) = \eta \cdot m(p,n), \tag{6}$$

where $m(p,n)$ is the number of new XY pairs created and $M(p,n)$ is the energy consumed. The pair creation is:

$$X(p,n+1) \leftarrow X(p,n+1) + m(p,n), \tag{7}$$

$$Y(p,n+1) \leftarrow Y(p,n+1) + m(p,n). \tag{8}$$

Explosion occurs only if $\mathcal{E}(p,n) \geq M(p,n)$; otherwise, m is reduced to the maximum affordable: $m(p,n) = \lfloor \mathcal{E}(p,n)/\eta \rfloor$. Leakage also applies: $L(p,n) = \lambda F(p,n)$.

Remark 2.11. The explosion count m in (5) grows linearly with the overshoot $F - C$, and each created pair costs energy η. This ensures that explosions are self-limiting: they drain the local energy reserve, reducing future explosion capacity.

2.8 Spatial Coupling

Excitations undergo nearest-neighbour diffusion on $\mathcal{G}$.

Definition 2.12 (Excitation diffusion)**.** After interaction and regime updates, excitations diffuse:

$$X(p,n+1) \leftarrow X(p,n+1) + D_X \sum_{q \in \mathcal{N}(p)} \left[X(q,n) - X(p,n)\right], \tag{9}$$

$$Y(p,n+1) \leftarrow Y(p,n+1) + D_Y \sum_{q \in \mathcal{N}(p)} \left[Y(q,n) - Y(p,n)\right]. \tag{10}$$

Values are rounded to the nearest non-negative integer. The constraint $D_X < 1/\deg_{\max}$ ensures that diffusion does not create negative counts in the continuous relaxation.

Remark 2.13. Diffusion is *conservative*: it redistributes excitations without creating or destroying them. Formally, $\sum_p X(p,n)$ is invariant under the diffusion step alone (up to rounding). The capacity energy $\mathcal{E}$ does *not* diffuse; it remains a local, immobile resource.

2.9 Bond Formation via Free-Energy Minimisation

We model bond formation as a local phase transition using a Landau free-energy functional [Landau, 1937].

Definition 2.14 (Order parameter)**.** The *bond order parameter* at vertex p is

$$\psi(p,n) = \sqrt{X(p,n) \cdot Y(p,n)},$$

measuring the geometric co-density of the two excitation species.

Definition 2.15 (Effective temperature)**.**

$$T_{\text{eff}}(p,n) = \frac{F(p,n)}{C}\,,$$

the normalised ripple intensity. High ripple corresponds to high effective temperature (disordered); low ripple to low temperature (ordered).

Definition 2.16 (Landau free energy)**.** The local *bond free energy* is

$$\mathcal{F}(\psi) = a_0\big(T_{\text{eff}} - T_c\big)\,\psi^2 + b\,\psi^4\,. \tag{11}$$

Bond formation is thermodynamically favourable when the free-energy gradient with respect to ψ is negative:

$$\frac{\partial \mathcal{F}}{\partial \psi} = 2\,a_0(T_{\text{eff}} - T_c)\,\psi + 4b\,\psi^3 < 0\,.$$

For $\psi > 0$ this requires

$$T_{\text{eff}}(p,n) < T_c - \frac{2b}{a_0}\,\psi^2(p,n)\,, \tag{12}$$

i.e., sufficiently low ripple (ordered local state) and sufficiently high co-density.

Definition 2.17 (Bond formation rate)**.** The number of bonds formed at vertex p is

$$B(p,n) = \begin{cases} \min\!\left(\left\lfloor \psi^2(p,n)\cdot \dfrac{|\partial\mathcal{F}/\partial\psi|}{b} \right\rfloor,\ \left\lfloor \dfrac{\mathcal{E}(p,n)}{\kappa} \right\rfloor,\ X(p,n),\ Y(p,n) \right) & \text{if (12) holds,} \\ 0 & \text{otherwise.} \end{cases} \tag{13}$$

Each bond:

- removes one X and one Y excitation from the free population,
- consumes energy κ from $\mathcal{E}(p,n)$,
- contributes $+\gamma_2$ to the structural state $S(p,n{+}1)$.

Bonds are *metastable*: they persist in S but obey the same ripple and energy rules as free excitations. In particular:

- A bonded vertex carries reduced capacity energy ($\mathcal{E}$ diminished by κ at formation).
- If a subsequent explosion at or near the bonded vertex raises F above $C + \Delta$, the bond may be disrupted: the structural contribution persists in S, but new excitations created by the explosion can overwhelm the local ordered phase.
- Bonds can therefore *form, persist, and later decay* through the same threshold mechanism that governs free excitations.

Remark 2.18. The quartic term $b\,\psi^4$ in (11) prevents unbounded bond formation. The condition $T_{\text{eff}} < T_c$ defines a genuine order–disorder transition: bonds form in the "ordered" (low ripple) phase and dissolve in the "disordered" (high ripple) phase. The fact that bonds are subject to the same dynamical rules as free excitations, rather than being permanent, is essential: it means the structural hierarchy (see section 5.2) is itself transient and history-dependent.

2.10 Structural State Evolution

Definition 2.19 (S update)**.**

$$S(p,n{+}1) = S(p,n) + \gamma_1\, N_{XY}(p,n) + \gamma_{XX}\, N_{XX}(p,n) + \gamma_2\, B(p,n)\,. \tag{14}$$

The structural state accumulates the history of all interactions (cross and self) and bonds. It has no decay term: once structure is created, it persists (though its influence via the ripple measure may diminish as the system freezes).

2.11 Complete Update Rule

The full time-step $n \to n{+}1$ proceeds in the following deterministic order:

1. **Interaction:** Sample $N_{XY}(p,n)$ and $N_{XX}(p,n)$ for all p per theorem 2.5. Annihilate excitations via (1)–(3).

2. **Ripple computation:** Compute $F(p,n)$ for all p.

3. **Regime classification and leakage/explosion:** Apply theorem 2.10. Create new excitations if explosive.

4. **Bond formation:** Compute ψ, T_{eff}, $\mathcal{F}$, and form bonds via theorem 2.17.

5. **Energy update:**

$$\mathcal{E}(p,n{+}1) = \mathcal{E}(p,n) - \underbrace{k_{XY}\cdot N_{XY}(p,n) + k_{XX}\cdot N_{XX}(p,n)}_{\text{interaction costs}} - \underbrace{L(p,n)}_{\text{leakage}} - \underbrace{M(p,n)}_{\text{explosion cost}} - \underbrace{\kappa\cdot B(p,n)}_{\text{bond cost}}, \tag{15}$$

   clamped to $\max(\cdot,0)$. Summing over all vertices, $\mathcal{E}_{\text{tot}}(n+1) \le \mathcal{E}_{\text{tot}}(n)$, with strict inequality whenever at least one site has positive cost at step n; see theorem 3.3.

6. **Structural update:** Apply (14).

7. **Diffusion:** Apply (9)–(10).

Definition 2.20 (Absorbing state)**.** A configuration Σ is *absorbing* if, for all $p \in \mathcal{V}$:

$$X(p)\cdot Y(p) = 0\,, \qquad X(p) \le 1 \;\; (\text{if } \alpha_{XX} > 0)\,, \qquad F(p) \le C\,.$$

The first two conditions ensure no cross- or self-interactions can occur; the third ensures no leakage or explosions. In an absorbing state $\Sigma(n) = \Sigma(n{+}1)$ for all subsequent n.

3 Main Results

Throughout this section, let $(\Sigma(n))_{n\ge 0}$ be the Markov chain defined by the update rule in section 2.11 on the filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_n)_{n\ge0}, \mathbb{P})$, where $\mathcal{F}_n = \sigma(\Sigma(0),\dots,\Sigma(n))$.

3.1 Well-Posedness

Proposition 3.1 (Well-posedness). *For any initial configuration $\Sigma(0)$ with $X(p,0), Y(p,0) \in \mathbb{N}_0$, $\mathcal{E}(p,0) \geq 0$, and $S(p,0) \in \mathbb{R}$ for all $p \in \mathcal{V}$, the update rule preserves these domains: for all $n \geq 0$,*

$$X(p,n) \in \mathbb{N}_0, \quad Y(p,n) \in \mathbb{N}_0, \quad \mathcal{E}(p,n) \geq 0, \quad S(p,n) \in \mathbb{R}.$$

Proof. Non-negativity of X and Y is ensured by capping N_{XY} and B at $\min(X,Y)$ and capping diffusion via $D < 1/\deg_{\max}$ with rounding to $\mathbb{N}_0$. Non-negativity of $\mathcal{E}$ is ensured by the clamp in step 5. S evolves by addition of non-negative terms and remains in $\mathbb{R}$. □

3.2 Energy Monotonicity

Definition 3.2 (Total depletion).

$$\mathcal{D}(n) = \sum_{p\in\mathcal{V}} \left[k_{XY}\, N_{XY}(p,n) + k_{XX}\, N_{XX}(p,n) + L(p,n) + M(p,n) + \kappa\, B(p,n)\right].$$

Theorem 3.3 (Energy Monotonicity). *The total capacity energy $\mathcal{E}_{\text{tot}}(n)$ satisfies:*

(a) $\mathcal{E}_{\text{tot}}(n+1) = \mathcal{E}_{\text{tot}}(n) - \mathcal{D}(n)$ *for all* $n \geq 0$.

(b) $\mathcal{D}(n) \geq 0$ *almost surely, hence* $\mathcal{E}_{\text{tot}}(n+1) \leq \mathcal{E}_{\text{tot}}(n)$ *a.s.*

(c) $\mathbb{E}[\mathcal{D}(n) \mid \mathcal{F}_n] > 0$ *whenever* $\Sigma(n)$ *is not absorbing.*

Consequently, $\left(\mathcal{E}_{\text{tot}}(n)\right)_{n\geq 0}$ *is a non-negative supermartingale.*

Proof. **(a)** Summing (15) over all $p \in \mathcal{V}$:

$$\mathcal{E}_{\text{tot}}(n+1) = \sum_{p} \mathcal{E}(p,n+1)$$
$$= \sum_{p}\Big[\mathcal{E}(p,n) - k_{XY}\, N_{XY}(p,n) - k_{XX}\, N_{XX}(p,n) - L(p,n) - M(p,n) - \kappa\, B(p,n)\Big] = \mathcal{E}_{\text{tot}}(n) - \mathcal{D}(n).$$

(The clamp to zero can only further reduce the sum, strengthening the inequality.)

(b) Each term in $\mathcal{D}(n)$ is a product of a positive constant and a non-negative count or rate. Hence $\mathcal{D}(n) \geq 0$.

(c) Suppose $\Sigma(n)$ is not absorbing. Then at least one of the following holds:

- There exists p with $X(p,n) \geq 1$ and $Y(p,n) \geq 1$. Then $R_{XY}(p,n) = \alpha_{XY}\, \omega_X \omega_Y\, X\, Y \geq \alpha_{XY}\omega_X\omega_Y > 0$, so $\mathbb{E}[k_{XY}\, N_{XY}(p,n)] > 0$.
- There exists p with $X(p,n) \geq 2$ and $\alpha_{XX} > 0$. Then $R_{XX}(p,n) \geq \alpha_{XX}\, \omega_X^2 > 0$, so $\mathbb{E}[k_{XX}\, N_{XX}(p,n)] > 0$.
- $F(p,n) > C$ for some p, giving $L(p,n) = \lambda\, F(p,n) > \lambda\, C > 0$.

In every case $\mathbb{E}[\mathcal{D}(n) \mid \mathcal{F}_n] > 0$.

Since $\mathcal{E}_{\text{tot}}(n) \geq 0$ and $\mathbb{E}[\mathcal{E}_{\text{tot}}(n+1) \mid \mathcal{F}_n] = \mathcal{E}_{\text{tot}}(n) - \mathbb{E}[\mathcal{D}(n) \mid \mathcal{F}_n] \leq \mathcal{E}_{\text{tot}}(n)$, the process $\left(\mathcal{E}_{\text{tot}}(n)\right)$ is a non-negative supermartingale. □

3.3 Finite Activity Bound

Theorem 3.4 (Finite Activity Bound)**.** *The total number of interactions, leakage events, explosion events, and bond formations across all space and time is almost surely finite. Specifically:*

$$\sum_{n=0}^{\infty} \sum_{p \in \mathcal{V}} N_{XY}(p,n) \;\le\; \frac{\mathcal{E}_{\mathrm{tot}}(0)}{k_{XY}} \qquad a.s. \tag{16}$$

More generally,

$$\sum_{n=0}^{\infty} \mathcal{D}(n) \le \mathcal{E}_{\mathrm{tot}}(0) \qquad a.s. \tag{17}$$

Proof. From theorem 3.3(a), telescoping:

$$\mathcal{E}_{\mathrm{tot}}(N) = \mathcal{E}_{\mathrm{tot}}(0) - \sum_{n=0}^{N-1} \mathcal{D}(n)\,.$$

Since $\mathcal{E}_{\mathrm{tot}}(N) \ge 0$ for all N:

$$\sum_{n=0}^{N-1} \mathcal{D}(n) \le \mathcal{E}_{\mathrm{tot}}(0)\,.$$

Taking $N \to \infty$ and using the monotone convergence theorem:

$$\sum_{n=0}^{\infty} \mathcal{D}(n) \le \mathcal{E}_{\mathrm{tot}}(0) < \infty\,.$$

Since $k_{XY}\, N_{XY}(p,n) \le \mathcal{D}(n)$ for each p, inequality (16) follows by extracting the cross-interaction term (with $k = k_{XY}$). Analogous bounds hold for X–X interactions ($\le \mathcal{E}_{\mathrm{tot}}(0)/k_{XX}$), total leakage ($\le \mathcal{E}_{\mathrm{tot}}(0)/\lambda C$), total explosions ($\le \mathcal{E}_{\mathrm{tot}}(0)/\eta$), and total bonds ($\le \mathcal{E}_{\mathrm{tot}}(0)/\kappa$). □

Corollary 3.5. *The system generates at most $\lfloor \mathcal{E}_{\mathrm{tot}}(0)/\min(k_{XY}, k_{XX}, \eta, \kappa) \rfloor$ total events across its entire history.*

3.4 Almost Sure Absorption

Theorem 3.6 (Almost Sure Absorption)**.** *There exists an almost surely finite random time $\tau < \infty$ such that $\Sigma(n)$ is absorbing for all $n \ge \tau$:*

$$\mathbb{P}\big(\exists\, \tau < \infty : \Sigma(n) \textit{ is absorbing } \forall\, n \ge \tau\big) = 1\,.$$

Proof. We argue that activity cannot persist indefinitely.

Step 1. By theorem 3.4, $\sum_n \mathcal{D}(n) < \infty$ a.s. Since $\mathcal{D}(n) \ge 0$, we have $\mathcal{D}(n) \to 0$ a.s.

Step 2. Suppose for contradiction that the system is non-absorbing for infinitely many n. At each such n, by the proof of theorem 3.3(c), there exists a vertex p_n where either:

(i) $X(p_n, n) \ge 1$ and $Y(p_n, n) \ge 1$, or

(ii) $F(p_n, n) > C$.

Case (i): If $X(p_n, n) \geq 1$ and $Y(p_n, n) \geq 1$, then $R_{XY}(p_n, n) \geq \alpha_{XY}\omega_X\omega_Y > 0$ and $\mathbb{P}(N_{XY} \geq 1) \geq 1 - e^{-\alpha_{XY}\omega_X\omega_Y} > 0$. When a cross-interaction occurs, $\mathcal{D}(n) \geq k_{XY} > 0$. If instead $X(p_n, n) \geq 2$ and $\alpha_{XX} > 0$, the same argument applies to N_{XX} with cost k_{XX}.

Since this positive-probability event occurs at infinitely many times, the Borel–Cantelli lemma (second form, noting conditional independence) gives $\sum_n \mathcal{D}(n) = \infty$ a.s. on the event that case (i) occurs infinitely often. This contradicts $\sum_n \mathcal{D}(n) < \infty$.

Case (ii): If $F(p_n, n) > C$ infinitely often, then $\mathcal{D}(n) \geq L(p_n, n) = \lambda\, F(p_n, n) > \lambda C > 0$ infinitely often, again contradicting $\mathcal{D}(n) \to 0$.

Step 3. Therefore, only finitely many non-absorbing steps occur. Let τ be the last such step; then $\Sigma(n)$ is absorbing for $n > \tau$. □

Remark 3.7 (Final configuration)**.** At absorption, the residual state consists of:

- residual excitations that are not co-located ($X \cdot Y = 0$ everywhere),
- residual capacity energy $\mathcal{E}(p, \tau) \geq 0$ (possibly positive),
- a frozen structural state $S(p, \tau)$ encoding the cumulative history of all interactions and bonds.

The final configuration depends on the entire stochastic history and is generically unique to each realisation. This *stochastic memory* of the last cascade is imprinted in the structural field.

Remark 3.8 (Why F_{avg} eventually decreases)**.** As total energy falls, total activity (interactions, explosions, bonds) must eventually fall. The increments to S therefore become smaller and less frequent. Since $F = |\Delta^2 S|$ is driven by those increments, their slowdown implies that the average ripple $F_{\text{avg}}(n)$ tends to decrease in the long run until $F \leq C$ everywhere and the system is quiescent.

3.5 Simulation

The dynamics were implemented in Python (see section A and the `model_simulation` folder). Figure 1 shows a typical run: total capacity energy decreases monotonically (theorem 3.3), excitations X and Y evolve asymmetrically until activity ceases, and the mean ripple F_{avg} rises then falls as structure builds and the system approaches absorption.

3.6 Convergence Rate

Proposition 3.9 (Expected absorption time bound)**.** *Let* $k_{\min} = \min(k_{XY}, k_{XX}, \lambda C, \eta, \kappa)$ *and define* $\delta = k_{\min} \cdot \alpha_{XY}\, \omega_X\omega_Y \cdot \min_{p:\, XY>0}(X(p,0)\, Y(p,0))$*. Then the expected time to absorption satisfies:*

$$\mathbb{E}[\tau] \;\leq\; \frac{\mathcal{E}_{\text{tot}}(0)}{\delta}\,.$$

Proof. While the system is non-absorbing, $\mathbb{E}[\mathcal{D}(n) \mid \mathcal{F}_n] \geq \delta > 0$ by the same argument as theorem 3.3(c). Since $\mathcal{E}_{\text{tot}}$ decreases by at least δ per step in expectation and is bounded below by zero, the expected number of non-absorbing steps is at most $\mathcal{E}_{\text{tot}}(0)/\delta$. □

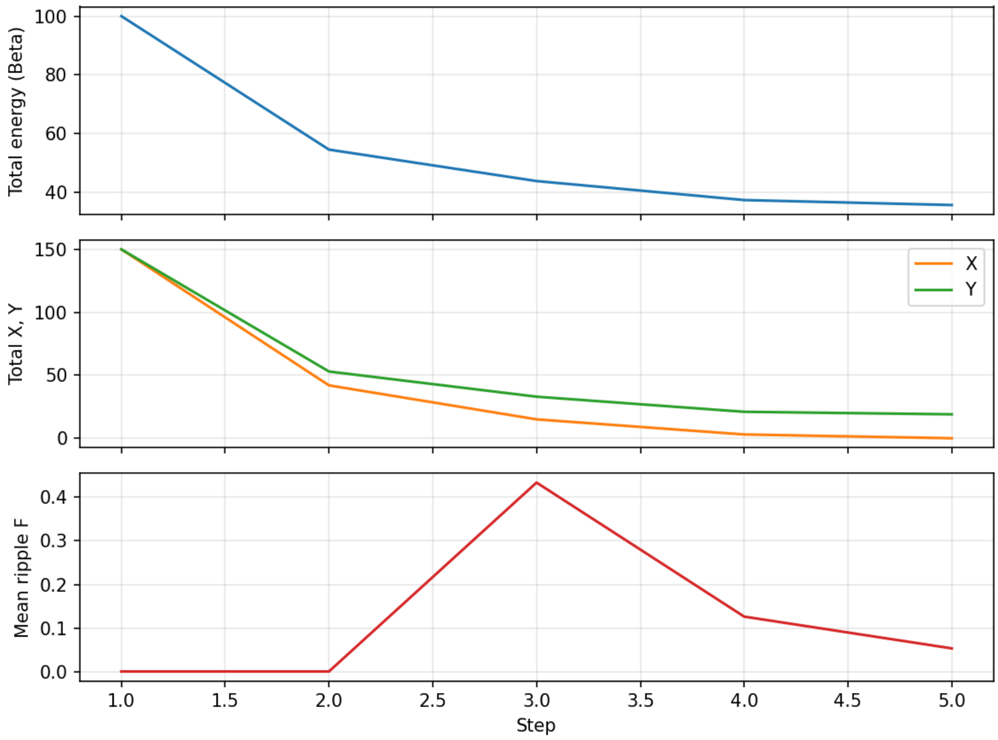

Figure 1: Simulation run on a periodic chain of $P = 50$ sites: initial $X = Y = 3$, $\mathcal{E}_{\text{tot}}(0) = 100$, default parameters. **Top:** total capacity energy (strictly decreasing). **Middle:** total X and Y (asymmetric depletion). **Bottom:** mean ripple F. Generated by `model_simulation/run_simulation.py`.

4 Phase Structure and Transient Complexity

4.1 Phase Diagram

The qualitative behaviour of the system depends on two dimensionless control parameters:

Definition 4.1 (Control parameters)**.**

$$\rho_0 = \frac{1}{P}\sum_p X(p,0)\,Y(p,0) \qquad \text{(initial co-density)}\,, \tag{18}$$

$$\varepsilon_0 = \frac{\mathcal{E}_{\text{tot}}(0)}{P} \qquad \text{(initial energy density)}\,. \tag{19}$$

Theorem 4.2 (Phase classification)**.** *The model exhibits three qualitative phases depending on* (ρ_0, ε_0)*:*

1. ***Immediate absorption*** *($\varepsilon_0 < k_{\min}$ or $\rho_0 = 0$, where $k_{\min} = \min(k_{XY}, k_{XX})$). Insufficient energy or excitations for sustained activity. The system absorbs within $O(1)$ steps.*

2. ***Dissipative decay*** *($\varepsilon_0 \geq k_{\min}$, $\rho_0 > 0$, initial $F < C + \Delta$ everywhere). Interactions occur and deplete energy, but no explosions are triggered. Excitation density decays monotonically. The structural state grows monotonically until absorption.*

3. ***Cascade regime*** *($\varepsilon_0 \gg k_{\min}$, ρ_0 sufficient to generate $F \geq C + \Delta$). Explosions create new excitation pairs, sustaining and amplifying activity in transient bursts. The system passes through a complexity peak before eventual absorption.*

Proof sketch. Phase 1 is immediate from the energy bound: if $\mathcal{E}(p,0) < k_{\min}$ for all p, no interaction can complete (each requires at least $k_{\min}$ energy), so $N_{XY} = N_{XX} = 0$ everywhere regardless of excitation counts.

Phase 2: without explosions, X and Y decrease monotonically (interactions and bonds only annihilate excitations). The number of interactions per step is bounded by $\min(X, Y)$, which is non-increasing. S grows by $\gamma_1 N_{XY} + \gamma_2 B \geq 0$.

Phase 3: explosions create excitations, which interact, which may increase F above the explosion threshold again, creating a cascade. But each cascade step costs energy (theorem 3.4), so the cascade is self-limiting. The total cascade duration is bounded by $\mathcal{E}_{\text{tot}}(0)/\eta$. □

4.2 Transient Complexity Peak

Definition 4.3 (Structural complexity)**.**

$$\mathcal{K}(n) = \frac{1}{P}\sum_{p\in\mathcal{V}} \left(S(p,n) - \bar{S}(n)\right)^2\,, \qquad \bar{S}(n) = \frac{1}{P}\sum_p S(p,n)\,.$$

This is the spatial variance of the structural state, measuring heterogeneity of accumulated structure across the graph.

Proposition 4.4 (Complexity peak). *In the cascade regime (theorem 4.2, Phase 3), the structural complexity $\mathcal{K}(n)$ satisfies:*

(a) $\mathcal{K}(0) = 0$ if $S(p, 0) = 0$ for all p.

(b) There exists $n^* > 0$ such that $\mathcal{K}(n^*) > 0$.

(c) $\mathcal{K}(n)$ is eventually constant (frozen at $\mathcal{K}(\tau)$).

Consequently, $\mathcal{K}$ has a non-trivial transient profile that first increases, may fluctuate, and ultimately freezes.

Proof. (a) is immediate. For (b), in the cascade regime at least one explosion occurs at some vertex p^* at some time n_1, adding excitations that produce $N_{XY}(p^*, n_1) > 0$ with positive probability. This increases $S(p^*, n_1+1)$ but not $S(q, n_1+1)$ for distant q, creating spatial variance. (c) follows from absorption: after τ, S is constant. □

4.3 Critical Threshold for Cascades

Proposition 4.5 (Cascade onset). *A necessary condition for the cascade regime is:*

$$\gamma_1 \, \alpha_{XY} \, \omega_X \omega_Y \, \rho_0 + \gamma_{XX} \, \alpha_{XX} \, \omega_X^2 \, \bar{X}_0 \, (\bar{X}_0 - 1)/2 > C\,,$$

where $\bar{X}_0$ is the mean initial X-count per vertex. That is, the expected structural increment from all interaction channels at initial density must exceed the ripple threshold.

Proof. For $F(p, n) > C$ at the earliest time $n = 2$, we need $|S(p, 2) - 2S(p, 1) + S(p, 0)| > C$. If $S(p, 0) = 0$, this requires $|S(p, 2) - 2S(p, 1)| > C$. Since $S(p, 1) = \gamma_1 \, N_{XY}(p, 0) + \gamma_{XX} \, N_{XX}(p, 0)$ and the expected contributions scale as $\gamma_1 \, \alpha_{XY} \omega_X \omega_Y \, \rho_0$ and $\gamma_{XX} \, \alpha_{XX} \omega_X^2 \, \bar{X}_0(\bar{X}_0 - 1)/2$ respectively, the expected magnitude of the second difference yields the stated condition. □

5 Emergent Properties and Interpretation

5.1 Layered Architecture

The model admits a natural decomposition into five conceptual layers, each emergent from the one below:

1. **Micro-field layer.** Stochastic X–Y and X–X interactions at individual vertices. The fundamental asymmetry ($\alpha_{YY} = 0$) and distinct frequencies $\omega_X \neq \omega_Y$ are the irreducible source of all dynamics.

2. **Energy layer.** Capacity energy $\mathcal{E}$ consumed per interaction. Provides the irreversible resource constraint.

3. **Instability control layer.** The ripple measure F and threshold C classify dynamics into quiescent, dissipative, and explosive regimes. This creates qualitative diversity from quantitative variation.

4. **Structural emergence layer.** Bond formation and structural state S create persistent, spatially heterogeneous configurations. These are the "fossils" of past activity.

5. **Thermodynamic arrow layer.** The monotonic decrease of $\mathcal{E}_{\text{tot}}$ imposes a global directionality on the dynamics. No time-reversal symmetry exists: the system has an intrinsic arrow.

5.2 Structural Hierarchy

The asymmetric interaction rules and bond formation produce a natural hierarchy of increasingly complex structures:

Level	Entity	Characteristics
0	Free excitation (X or Y)	Single species, mobile, reactive
1	Co-located pair (X + Y at same p)	Can interact, annihilate, or bond
2	Bond (X–Y composite in S)	Metastable, reduced $\mathcal{E}$, obeys same rules
3	Clump (cluster of bonds on $\mathcal{G}$)	Spatially extended, collective ripple

Each level emerges from the dynamics of the level below. Level-0 entities are placed by initial conditions and explosions. Level-1 co-locations arise from diffusion. Level-2 bonds form when the free-energy condition (12) is satisfied. Level-3 clumps form when neighbouring vertices independently satisfy the bonding condition, creating spatially correlated structural states.

The hierarchy is *transient*: clumps form, persist while local energy supports them, and ultimately freeze when the system absorbs. The frozen final state is not optimised, not symmetric, and not predictable from initial conditions alone; it is simply the arrested residue of the last cascade.

5.3 Thermodynamic Arrow

Corollary 5.1 (Irreversibility)**.** *The dynamics are time-irreversible. For any non-absorbing state Σ, the probability of returning to a state with higher total energy is exactly zero:*

$$\mathbb{P}\big(\mathcal{E}_{\text{tot}}(n+1) > \mathcal{E}_{\text{tot}}(n)\big) = 0 \qquad \forall\, n \geq 0\,.$$

This follows immediately from (15) and the non-negativity of all depletion terms. The system possesses a *strict thermodynamic arrow* without any need to invoke entropy or ergodic assumptions.

5.4 Stochastic Memory

The final frozen structural state $S(p, \tau)$ depends on the entire realisation of the stochastic process. Different runs from the same initial conditions produce generically different final configurations. This "stochastic memory" is a permanent record of the system's particular history, a discrete analogue of symmetry breaking in equilibrium phase transitions.

The absorbing configuration is not optimised with respect to any functional, not symmetric under any non-trivial automorphism of $\mathcal{G}$, and not reconstructible from the initial conditions alone. It is the arrested residue of the last cascade: a fossil record of the system's entire stochastic biography, frozen into the structural field when the energy ran out.

5.5 Self-Limiting Cascades

In the explosive regime, cascades are necessarily self-limiting. Each explosion creates new excitations (increasing future interaction potential) but costs energy (decreasing future capacity). The cascade terminates when either:

- local energy is exhausted ($\mathcal{E}(p) < \eta$), or
- created excitations diffuse and dilute below the interaction threshold.

This produces burst-like activity with heavy-tailed duration distributions, reminiscent of avalanche dynamics in SOC models, but guaranteed to eventually cease entirely.

6 Discussion

6.1 Relation to Existing Frameworks

Contact process and directed percolation. The contact process on a graph has a single species that can infect neighbours or recover. It exhibits a continuous phase transition between an absorbing state and an active steady state. Our model differs in four respects: (i) two *asymmetric* species with distinct frequencies and interaction rules, (ii) a global energy constraint that makes absorption *certain* rather than a phase boundary, (iii) structural memory that breaks the Markov property of the order parameter alone, and (iv) a structural hierarchy (excitation $\to$ pair $\to$ bond $\to$ clump) absent from single-species models.

Sandpile models. The Abelian sandpile [Dhar, 1990] achieves criticality through slow external driving and fast relaxation (separation of timescales). Our model has no external drive; it runs on a finite endowment. The transient complexity peak (theorem 4.4) is analogous to the SOC steady state, but is inherently non-stationary and terminal.

Reaction–diffusion annihilation. The $A + B \to \varnothing$ system [Toussaint & Wilczek, 1983] on lattices shows anomalous kinetics (density decay $\sim t^{-d/4}$ in d dimensions). Our model extends this by adding an energy field, threshold-driven pair creation ($\varnothing \to A + B$ at a cost), and structural accumulation.

Landau theory in non-equilibrium systems. The use of a Landau free energy for bond formation (section 2.9) imports equilibrium phase-transition phenomenology into a far-from-equilibrium, transient context. This is justified because bond formation is a local, fast process relative to the global energy depletion timescale, allowing a quasi-static approximation at the bond level.

6.2 Numerical illustration: Big Bang analogue

With hot, dense initial conditions (high $\mathcal{E}_{\text{tot}}(0)$ and excitation densities), a single run exhibits a burst of activity, growth of structure S, and then irreversible decay toward absorption: a runnable analogue of a "Bang" followed by expansion and cooling. Figure 2 shows total energy, total X and Y, total structure, and mean ripple over time. The code is `model_simulation/run_bigbang.py`.

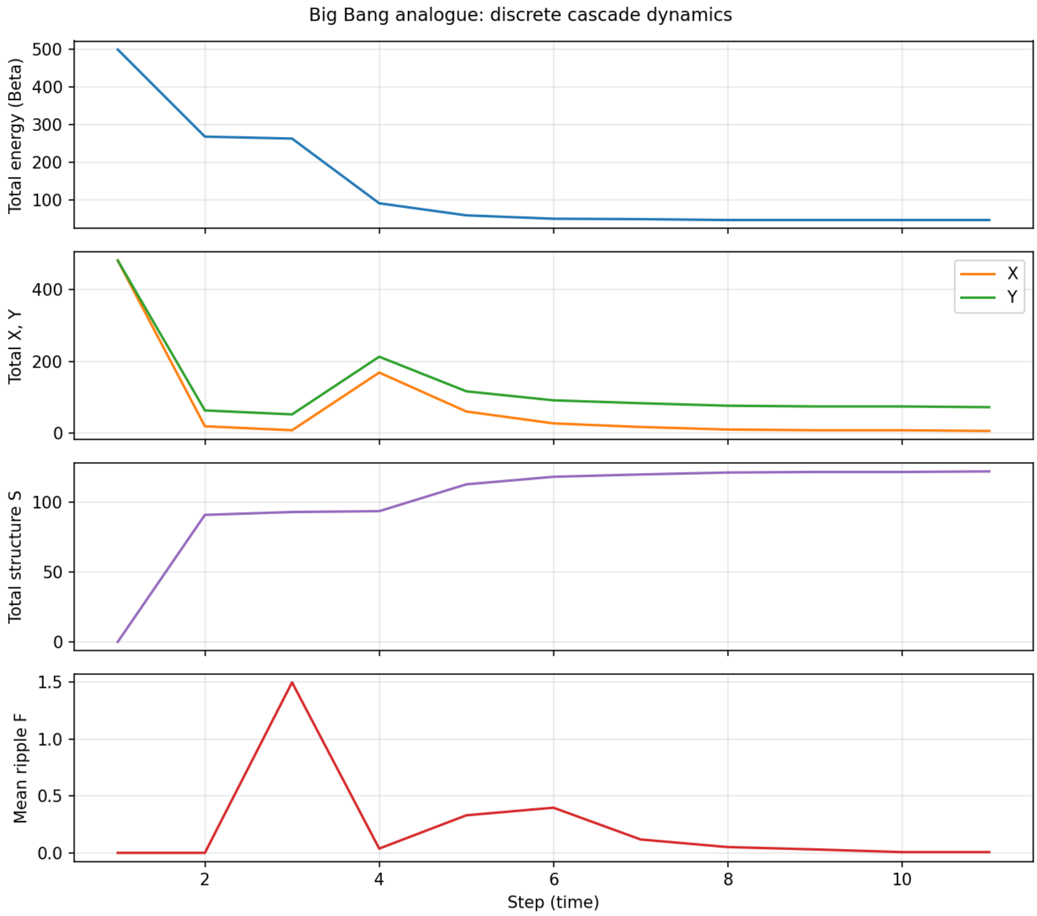

Figure 2: Big Bang analogue run: $P = 100$ sites, high initial energy and X, Y. **Top to bottom:** total energy (monotone decrease), total X and Y (asymmetric evolution), total structure S, mean ripple F.

A second analogue is an *early distribution to current universe* run: initial conditions are concentrated in a "primordial" region (left fraction of the chain); the rest of space is empty. Diffusion and dynamics then spread excitations and energy across the graph until absorption. Figure 3 shows the global time series (energy, X, Y, S, F); Figure 4 shows the spatial distribution at several times: expansion from the hot dense region into the rest of the "universe." The code is `model_simulation/run_cosmology.py`.

A third narrative is the *quantum fluctuation* run: a single small region is given a little X, Y, and energy; the rest of the graph is empty. The fluctuation triggers the same cascade dynamics; the burst and subsequent decay are interpreted as a "quantum" trigger of a Bang-like evolution (fig. 5). The code is `model_simulation/run_quantum_bang.py`; the interpretation is unproven but illustrates that the model admits a minimal-seed initial condition.

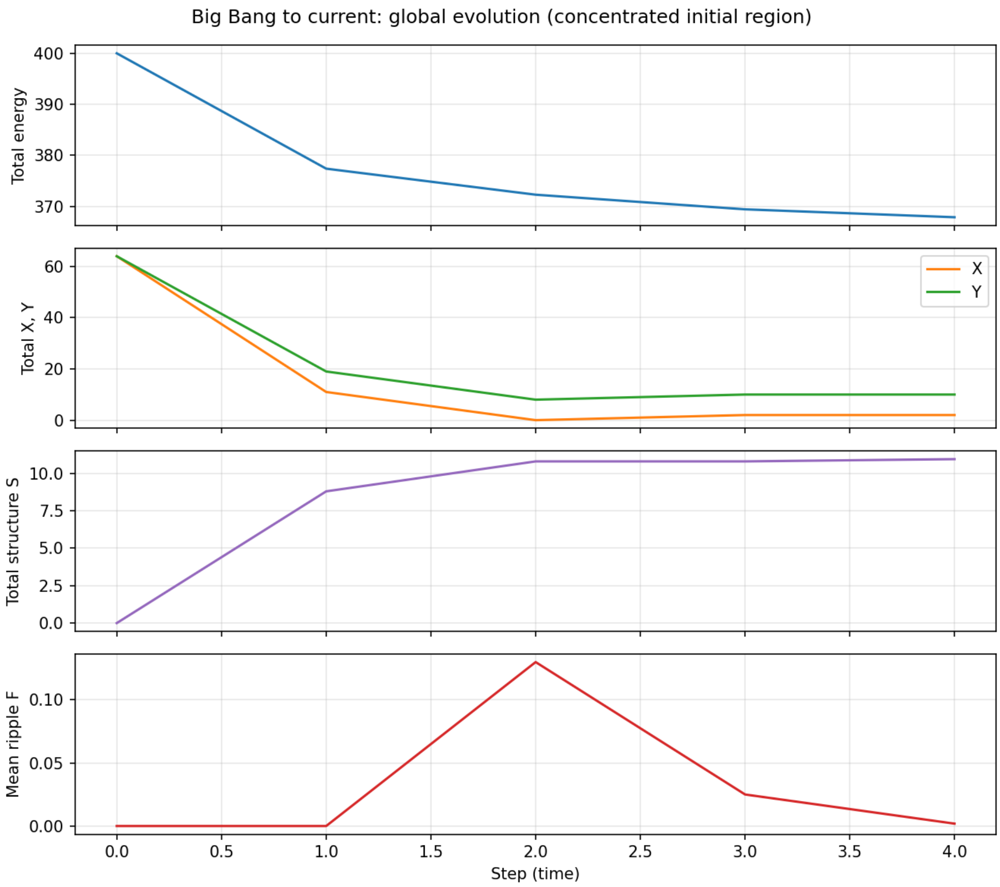

Figure 3: Early-universe to current analogue: global evolution. Total energy (top), total X and Y (middle), total structure S and mean ripple F (bottom) vs. time. Same run as fig. 4.

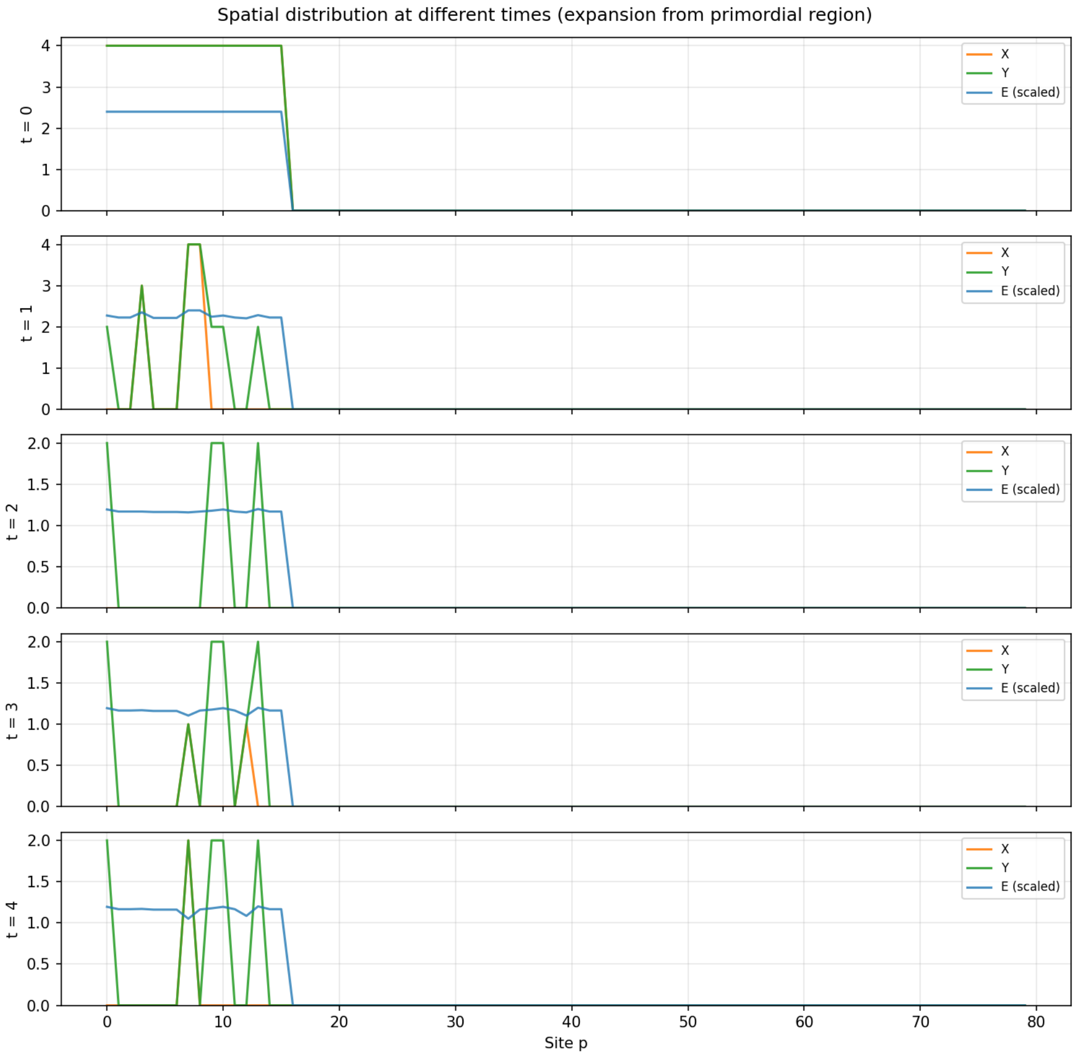

Figure 4: Early-universe to current analogue: spatial distribution of X, Y, and energy (scaled) at several time steps. Initial state is concentrated in the left 25% of sites; diffusion and dynamics spread the activity until absorption.

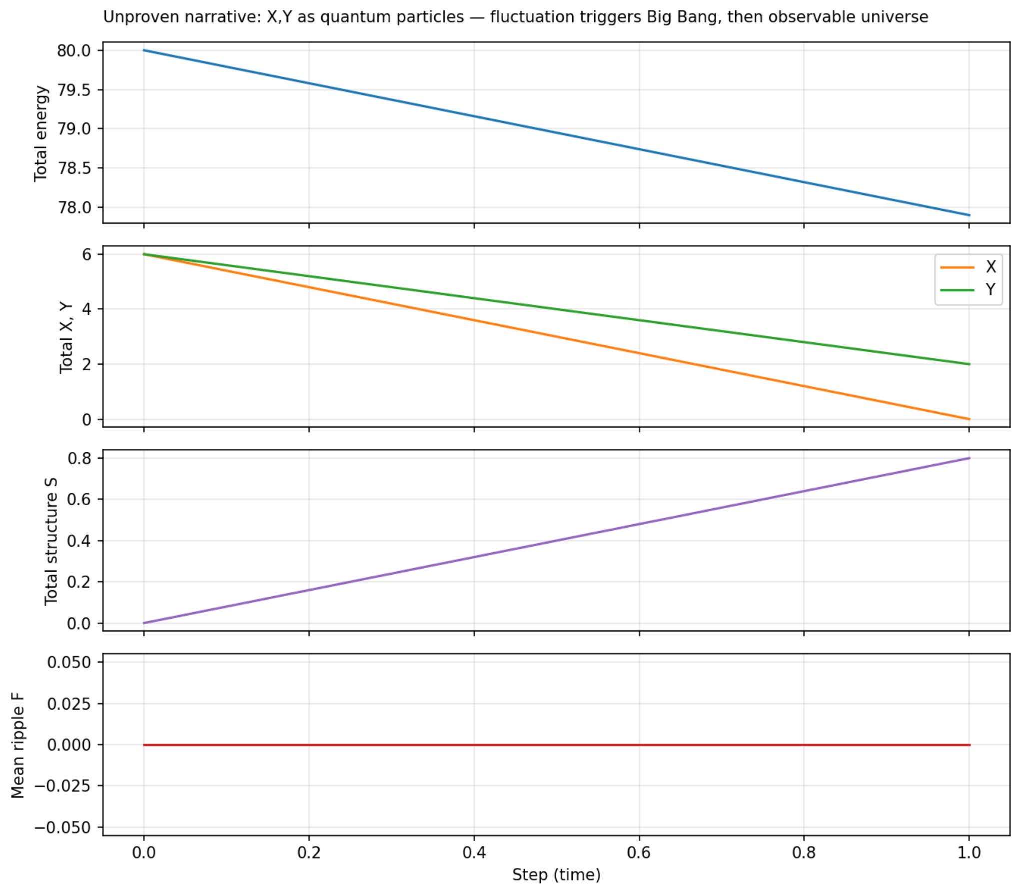

Figure 5: Quantum fluctuation run: one small region with X, Y, and energy; rest empty. **Top to bottom:** total energy, total X and Y (asymmetric outcome), total structure S, mean ripple F. Narrative: the fluctuation triggers the cascade (unproven).

6.3 Open Problems

1. **Universality class.** Does the absorbing-state transition exhibit directed percolation universality, or does the energy constraint define a new universality class?

2. **Cascade size distribution.** What is the distribution of cascade sizes (total excitations created per explosive event)? Preliminary analysis suggests a truncated power law, but rigorous results are lacking.

3. **Structural complexity scaling.** How does the peak structural complexity $\max_n \mathcal{K}(n)$ scale with graph size P and initial energy density ε_0?

4. **Infinite-volume limit.** Can the model be extended to $\mathbb{Z}^d$ with appropriate initial conditions? The finite energy constraint may need to be replaced by a density constraint for this limit to be non-trivial.

5. **Continuum limit.** Under appropriate rescaling of space, time, and field variables, does the model converge to a stochastic PDE? Candidates include reaction–diffusion equations with a depleting source term.

6. **Metastable bond lifetimes.** What is the expected lifetime of a bonded configuration before disruption by a nearby explosion? This connects to nucleation theory in statistical mechanics.

7. **Role of asymmetry** (resolved in Part II Goswami [2026b]). What happens in the symmetric limit $\alpha_{XX} = \alpha_{YY}$, $\omega_X = \omega_Y$? Does the structural hierarchy collapse? The forbidden Y–Y channel and the frequency mismatch are constitutive assumptions; understanding their necessity (or relaxation) is an important structural question. *Part II proves that both asymmetries are independently necessary: symmetry causes the hierarchy to collapse, the forbidden Y–Y channel is required for bond formation, and the frequency mismatch is required for non-trivial dynamics.*

6.4 Potential Extensions

- **Multiple excitation species.** Generalise from two species to K species with species-specific interaction rules and energy costs.
- **Adaptive graph topology.** Allow the graph $\mathcal{G}$ itself to evolve, adding edges where bonds form, removing edges where energy is depleted.
- **External driving.** Add a slow energy injection term to study the transition from guaranteed absorption to possible steady-state behaviour (connecting to SOC models).
- **Spatial energy transport.** Allow $\mathcal{E}$ to diffuse, creating energy gradients and directed flows.

7 Conclusion

We have introduced a discrete stochastic cascade model on finite graphs that couples four state variables (two excitation species, a capacity energy field, and a structural state) through layered update rules with three dynamical regimes. The model is fully specified by twelve parameters and a choice of graph topology.

The central structural result is that the total capacity energy is a strict Lyapunov function (non-negative supermartingale) for the dynamics: it decreases at every active time step and cannot increase. This single invariant implies finite total activity, almost sure absorption, and a strict thermodynamic arrow, all as rigorous consequences, not assumptions.

The model generates rich transient phenomena. In the cascade regime, explosive pair creation amplifies activity in self-limiting bursts, building spatially heterogeneous structural configurations before the system freezes into a unique, history-dependent absorbing state. Bond formation is governed by a Landau free-energy functional that embeds a local order–disorder phase transition within the global irreversible dynamics.

The framework provides a minimal, internally consistent setting for studying the interplay between stochastic interactions, irreversible resource depletion, threshold-driven instabilities, and emergent structure. It sits at the intersection of interacting particle systems, absorbing-state phase transitions, and reaction–diffusion dynamics, and raises several open questions regarding universality, scaling, and continuum limits that merit further investigation.

A Simulation Pseudocode

B Parameter Sensitivity

The qualitative behaviour is controlled by a small number of dimensionless ratios:

Ratio	Controls	Effect of increasing
$\alpha_{XY}\omega_X\omega_Y\,\rho_0/C$	Cascade onset	Easier to trigger explosions
α_{XX}/α_{XY}	Self- vs. cross-interaction	More X depletion, fewer bonds
ω_X/ω_Y	Frequency asymmetry	Stronger species differentiation
$\mathcal{E}_{\text{tot}}(0)/(k_{XY}\,P\,\rho_0)$	System lifetime	Longer active phase
η/k_{XY}	Explosion cost ratio	Fewer but costlier explosions
$D_X\ \deg_{\max}$	Diffusion strength	Faster spatial mixing
T_c	Bond formation ease	More bonds at higher T_c
κ/k_{XY}	Bond vs. interaction cost	Relative bond preference

C Key Equations Summary

For quick reference, the main equations of the model are collected below.

- **Interaction rates:** $R_{XY} = \alpha_{XY}\,\omega_X\,\omega_Y\,X\,Y$, $R_{XX} = \alpha_{XX}\,\omega_X^2\,X(X-1)/2$; Y–Y is forbidden ($\alpha_{YY} = 0$).
- **Ripple:** $F(p,n) = |S(p,n) - 2\,S(p,n{-}1) + S(p,n{-}2)| = |\Delta^2 S(p,n)|$.
- **Leakage:** $L = \lambda\,F$ when $C < F < C + \Delta$.
- **Explosion:** $m = \lfloor (F - C)/\Delta \rfloor$, $M = \eta\,m$ when $F \geq C + \Delta$.
- **Energy update:** $\mathcal{E}(p,n{+}1) = \mathcal{E}(p,n) - k_{XY}N_{XY} - k_{XX}N_{XX} - L - M - \kappa B$, clamped to ≥ 0.

- **Structural update:** $S(p, n+1) = S(p, n) + \gamma_1 N_{XY} + \gamma_{XX} N_{XX} + \gamma_2 B$.
- **Global invariant:** $\mathcal{E}_{\text{tot}}(n+1) = \mathcal{E}_{\text{tot}}(n) - \mathcal{D}(n)$, $\mathcal{D}(n) \geq 0$; strict decrease when active.

References

P. Bak, C. Tang, and K. Wiesenfeld. Self-organized criticality: An explanation of the $1/f$ noise. *Physical Review Letters*, 59(4):381–384, 1987.

D. Dhar. Self-organized critical state of sandpile automaton models. *Physical Review Letters*, 64(14):1613–1616, 1990.

T. E. Harris. Contact interactions on a lattice. *The Annals of Probability*, 2(6):969–988, 1974.

M. Henkel, H. Hinrichsen, and S. Lübeck. *Non-Equilibrium Phase Transitions, Volume 1: Absorbing Phase Transitions*. Springer, 2008.

H. Hinrichsen. Non-equilibrium critical phenomena and phase transitions into absorbing states. *Advances in Physics*, 49(7):815–958, 2000.

L. D. Landau. On the theory of phase transitions. *Zh. Eksp. Teor. Fiz.*, 7:19–32, 1937. [Reprinted in *Collected Papers of L.D. Landau*, Pergamon, 1965].

B. P. Lee. Renormalization group calculation for the reaction $kA \to \varnothing$. *Journal of Physics A*, 27(8):2633–2652, 1994.

T. M. Liggett. *Stochastic Interacting Systems: Contact, Voter and Exclusion Processes*. Springer, 1999.

S. Goswami. Discrete stochastic cascade dynamics on finite graphs with irreversible energy depletion. Part II: Necessity of species asymmetry for structural hierarchy. *Submitted*, 2026.

D. Toussaint and F. Wilczek. Particle–antiparticle annihilation in diffusive motion. *The Journal of Chemical Physics*, 78(5):2642–2647, 1983.

Algorithm 1 Discrete cascade model. $\mathcal{E}(p)$ denotes capacity energy. Note the asymmetry: X–X self-interaction is computed; Y–Y is absent.

Require: Graph $\mathcal{G} = (\mathcal{V}, \mathcal{A})$, parameters, initial fields $X(p)$, $Y(p)$, $\mathcal{E}(p)$, $S(p)$ for all $p \in \mathcal{V}$

for $n = 0, 1, 2, \ldots$ **do**
    $active \leftarrow$ FALSE

    for each vertex $p \in \mathcal{V}$ **do** ▷ Local updates
        Cross-interaction (X–Y)
        $R_{XY} \leftarrow \alpha_{XY}\, \omega_X\, \omega_Y\, X(p)\, Y(p)$
        $N_{XY} \leftarrow \min\big(\mathrm{Poi}(R_{XY}),\ X(p),\ Y(p)\big)$
        $X(p) \leftarrow X(p) - N_{XY};\quad Y(p) \leftarrow Y(p) - N_{XY}$
        Self-interaction (X–X); Y–Y forbidden
        $R_{XX} \leftarrow \alpha_{XX}\, \omega_X^2\, X(p)\big(X(p) - 1\big)/2$
        $N_{XX} \leftarrow \min\big(\mathrm{Poi}(R_{XX}),\ \lfloor X(p)/2 \rfloor\big)$
        $X(p) \leftarrow X(p) - 2\, N_{XX}$
        Ripple
        if $n \geq 2$ **then**
            $F \leftarrow \big|S(p, n) - 2\, S(p, n{-}1) + S(p, n{-}2)\big|$
        else
            $F \leftarrow 0$
        end if
        Regime classification and explosion
        $L \leftarrow 0;\quad M \leftarrow 0;\quad m \leftarrow 0$
        if $C < F < C + \Delta$ **then**
            $L \leftarrow \lambda\, F$
        else if $F \geq C + \Delta$ **then**
            $L \leftarrow \lambda\, F$
            $m \leftarrow \lfloor (F - C)/\Delta \rfloor;\quad M \leftarrow \eta\, m$
            if $M > \mathcal{E}(p)$ **then**
                $m \leftarrow \lfloor \mathcal{E}(p)/\eta \rfloor;\quad M \leftarrow \eta\, m$
            end if
            $X(p) \leftarrow X(p) + m;\quad Y(p) \leftarrow Y(p) + m$
        end if
        Bond formation (Landau free energy)
        $\psi \leftarrow \sqrt{X(p) \cdot Y(p)};\quad T_{\mathrm{eff}} \leftarrow F/C$
        $\partial \mathcal{F}/\partial \psi \leftarrow 2\, a_0 (T_{\mathrm{eff}} - T_c)\, \psi + 4b\, \psi^3$
        $B \leftarrow 0$
        if $\partial \mathcal{F}/\partial \psi < 0$ **and** $\psi > 0$ **then**
            $B \leftarrow \min\big(\lfloor \psi^2 |\partial \mathcal{F}/\partial \psi| / b \rfloor,\ \lfloor \mathcal{E}(p)/\kappa \rfloor,\ X(p),\ Y(p)\big)$
            $X(p) \leftarrow X(p) - B;\quad Y(p) \leftarrow Y(p) - B$
        end if
        Energy and structural updates
        $cost \leftarrow k_{XY}\, N_{XY} + k_{XX}\, N_{XX} + L + M + \kappa\, B$
        $\mathcal{E}(p) \leftarrow \max\big(\mathcal{E}(p) - cost,\ 0\big)$
        $S(p) \leftarrow S(p) + \gamma_1\, N_{XY} + \gamma_{XX}\, N_{XX} + \gamma_2\, B$
        if $N_{XY} + N_{XX} + L + M + B > 0$ **then**
            $active \leftarrow$ TRUE
        end if
    end for

    Diffusion (conservative: energy does not diffuse)